

Maven Project To Connect To Web Servers Using Selenium Dependency

Step1:

To create a basic Maven Java project, run the following command:

```
$mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp05 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
```

```
vaishnav@vaishnavi-VirtualBox: ~$ cd ..
vaishnav@vaishnavi-VirtualBox: ~$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenSeleniumApp03 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.apache.maven:standalone-pom >-----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] [ pom ]-----
[INFO]
[INFO] >>> mvn:archetype-plugin:1.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO] <<< mvn:archetype-plugin:1.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO]
[INFO] --- mvn:archetype-plugin:1.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] Downloading from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml
[INFO] Downloaded from central: https://repo.maven.apache.org/maven2/archetype-catalog.xml (18 MB at 7.3 MB/s)
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] Parameter: basedir, Value: /home/vaishnavi/devops21
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenSeleniumApp03
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] Project created from Old (1.x) Archetype in dir: /home/vaishnavi/devops21/MyMavenSeleniumApp03
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 4.456 s
[INFO] Finished at: 2025-03-17T08:58:39+05:30
[INFO]
vaishnavi@vaishnavi-VirtualBox: ~$ cd MyMavenSeleniumApp03
vaishnavi@vaishnavi-VirtualBox: ~$ cd MyMavenSeleniumApp03$ ls
pom.xml
```

The command can be broken down as:

DgroupId=com.example – project’s base package (reverse-domain style).

-DartifactId=my-maven-app – defines the name of the project (will also be the folder name).

-DarchetypeArtifactId=maven-archetype-quickstart – sets the Maven template to be the basic Java application.

-DinteractiveMode=false – deactivates interactive mode and uses the given values automatically.

Step 2:

Use \$cd command to go into the directory.

Execute \$tree command to view the structure of the project

Step 3:

Edit the pom.xml file

Add the required dependencies and plugins required from central maven repository

For given project, we will download unit testing dependency from Jenkins.

```
<project xmlns="http://maven.apache.org/POM/4.0.0" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance"
xsi:schemaLocation="http://maven.apache.org/POM/4.0.0 http://maven.apache.org/maven-v4_0_0.xsd">
<modelVersion>4.0.0</modelVersion>
<groupId>com.example</groupId>
<artifactId>MyMavenSeleniumApp05</artifactId>
<packaging>jar</packaging>
<version>1.0-SNAPSHOT</version>
<name>MyMavenSeleniumApp05</name>
<url>http://maven.apache.org</url>
<dependencies>
<dependency>
<groupId>junit</groupId>
<artifactId>junit</artifactId>
<version>3.8.1</version>
<scope>test</scope>
</dependency>
<!-- https://mvnrepository.com/artifact/org.seleniumhq.selenium/selenium-java -->
<dependency>
<groupId>org.seleniumhq.selenium</groupId>
<artifactId>selenium-java</artifactId>
<version>4.29.0</version>
</dependency>
</dependencies>
<!-- Build: Configuring plugins and build settings -->
<build>
<plugins>
```

<!-- Example: Maven Compiler Plugin to compile Java code -->

<plugin>

<groupId>org.apache.maven.plugins</groupId>

<artifactId>maven-compiler-plugin</artifactId>

<version>3.8.1</version>

<configuration>

<source>11</source>

<target>11</target>

</configuration>

</plugin>

<plugin>

<groupId>org.apache.maven.plugins</groupId>

<artifactId>maven-shade-plugin</artifactId>

<version>3.2.4</version>

<executions>

<execution>

<phase>package</phase>

<goals>

<goal>shade</goal>

</goals>

<configuration>

<transformers>

<transformer

implementation="org.apache.maven.plugins.shade.resource.ManifestResourceTransformer">

<mainClass>com.example.App</mainClass>

</transformer>

</transformers>

</configuration>

</execution>

</executions>

</plugin>

```
<!-- Example: Maven Surefire Plugin to run tests -->
<plugin>
<groupId>org.apache.maven.plugins</groupId>
<artifactId>maven-surefire-plugin</artifactId>
<version>2.22.2</version>
</plugin>
<plugin>
    <groupId>org.apache.maven.plugins</groupId>

    <artifactId>maven-jar-plugin</artifactId>
    <version>3.0.1</version>
    <configuration>
        <archive>
            <manifestEntries>
                <Main-Class>com.example.App</Main-Class>
            </manifestEntries>
        </archive>
    </configuration>
</plugin>
</plugins>
</build>
</project>
```

Step 4:

Execute the command `$mvn compile`

If the command shoes output as “build success”, it means that the project has compiled successfully

```

[INFO] Scanning for projects...
[INFO]
[INFO] -----< com.example:MyHavenApp >-----
[INFO] Building MyHavenApp 1.0-SNAPSHOT
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyHavenApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is plat
form dependent!
[INFO] skip non existing resourceDirectory /home/vaishnavi/devops21/MyHavenApp/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.8.1:compile (default-compile) @ MyHavenApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 0.713 s
[INFO] Finished at: 2026-03-24T21:20:16+05:30
[INFO]

```

Step 5:

- Right-click on the webpage and select Inspect.
- The HTML code will be displayed. Click on the cursor icon (arrow symbol) available on the top-left of the inspect panel, then hover over the username field to identify its ID, which is user-name.
- Similarly, check the search button (ID:submit_search). Make a note of these IDs, as they will be used in the App.java file.
- Next, Edit the App.java file

```
package com.example;
```

```
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.firefox.FirefoxDriver;
import org.openqa.selenium.firefox.FirefoxOptions;
```

```
public class App {
    public static void main(String[] args) {
```

```
        // 🚀 Setup Firefox for Jenkins (Headless mode)
        FirefoxOptions options = new FirefoxOptions();
        options.addArguments("--headless");           // REQUIRED
        options.addArguments("--no-sandbox");         // Recommended
        options.addArguments("--disable-dev-shm-usage"); // Recommended
```

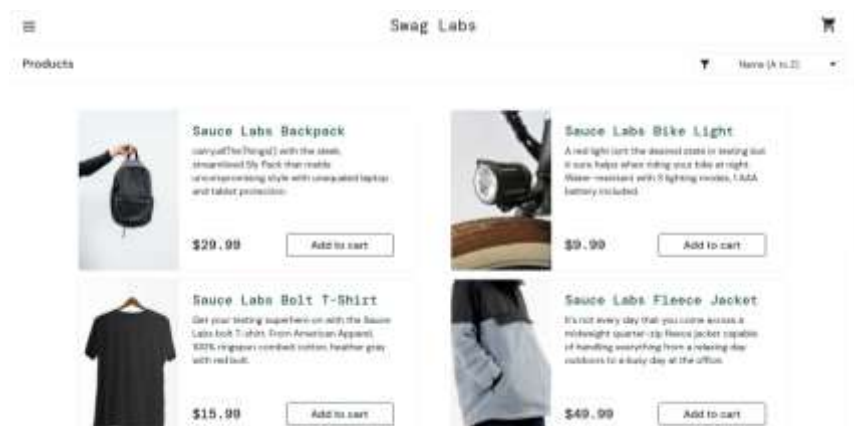
```
        WebDriver driver = new FirefoxDriver(options);
```

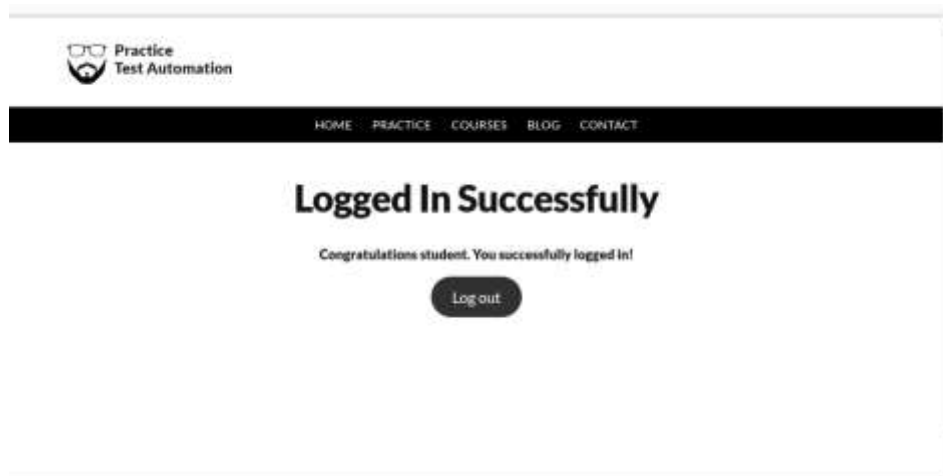
```
    try {
        // ---- Test 1: SauceDemo Login ----
        driver.get("https://www.saucedemo.com/");
        driver.findElement(By.id("user-name")).sendKeys("standard_user");
        driver.findElement(By.id("password")).sendKeys("secret_sauce");
        driver.findElement(By.id("login-button")).click();
    }
```


Execute `$ java -jar target/MyMavenSeleniumApp03-1.0-SNAPSHOT`

If we get error in executing the above jar file do run the below command

```
$mvn exec:java -Dexec.mainClass="com.example.App"
```

[illegible]



Pushing The Project into git

Execute the commands:

```
$ git config --global user.name "vaishnavim1121"
```

```
$ git config --global user.email "vaishnavim1121@gmail.com"
```

```
$git init
```

```
$git add .
```

```
$git commit -m "initial commit"
```

```
$ git remote add origin https://github.com/vaishnavim1121/MyMavenSeleniumFirefox.git
```


\$ git remote set-url origin https://*PAT*@github.com/vaishnavim1121/MyMavenSeleniumFirefox

\$ git push -u origin master

```

Initialized empty Git repository in /home/vaishnavi/Desktop/MyMavenSelenium/.git/
vaishnavi@vaishnavi-VirtualBox: /home/vaishnavi/Desktop/MyMavenSelenium$ git add .
vaishnavi@vaishnavi-VirtualBox: /home/vaishnavi/Desktop/MyMavenSelenium$ git commit -m "First commit"
[master (root-commit) 306b187] first commit
 33 files changed, 258 insertions(+)
 create mode 100644 pom.xml
 create mode 100644 src/main/java/com/example/App.java
 create mode 100644 src/test/java/com/example/AppTest.java
 create mode 100644 target/maven-selenium/target-1.6-SNAPSHOT.jar
 create mode 100644 target/classes/com/example/App.class
 create mode 100644 target/maven-archiver/pom.properties
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/inputFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst
 create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
 create mode 100644 target/surefire-reports/TEST-com.example.AppTest.xml
 create mode 100644 target/surefire-reports/com.example.AppTest.txt
 create mode 100644 target/test-classes/com/example/AppTest.class
vaishnavi@vaishnavi-VirtualBox: /home/vaishnavi/Desktop/MyMavenSelenium$ git remote add origin https://github.com/vaishnavim1121/MyMavenSeleniumApp04.git
vaishnavi@vaishnavi-VirtualBox: /home/vaishnavi/Desktop/MyMavenSelenium$ git remote set-url origin https://ghp_9Wp5Gxp2Q6FO7naCyrja6ZV9mTf10g/github.com/vaishnavim1121/MyMavenSeleniumApp04
vaishnavi@vaishnavi-VirtualBox: /home/vaishnavi/Desktop/MyMavenSelenium$ git push -u origin master
Enumerating objects: 39, done.
Counting objects: 100% (38/39), done.
Delta compression using up to 10 threads
Compressing objects: 100% (38/38), done.
Writing objects: 100% (39/39), 8.87 KiB | 1.79 MiB/s, done.
Total 39 (delta 0), reused 8 (delta 0), pack-reused 0
To https://github.com/vaishnavim1121/MyMavenSeleniumApp04
 * [new branch]      master -> master
branch 'master' set up to track remote branch 'master' from 'origin'

```